Learning Objective 3.4: I can calculate the phase weights required to steer a beam to a given angle and predict the resulting array pattern.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **From geometry to phase.** The course array is a uniform linear array of $N = 8$ patch elements spaced $d = 14$ mm apart, receiving at $f = 10.3$ GHz. A plane wave arrives $\theta_0 = 25^\circ$ from broadside.

   (a) In one or two sentences, explain which element the wavefront reaches first and why a fixed phase applied to each element can make all eight received signals add together.

   (b) Compute the wavelength, the electrical spacing d/λ, and the product kd in degrees.

   kd =

   (c) Compute the progressive element-to-element phase $\Delta\phi$ required to steer the beam to 25°.

   $\Delta\phi$ =

   (d) Compute the arrival-time difference between neighboring elements at this steer angle, in picoseconds.

   Δt =

2. **The phase table.** We command the same array to $\theta_0 = -40^\circ$. Element 0 is the phase reference, and the ADAR1000 accepts a phase setting between 0° and 360°.

   (a) Compute $\Delta\phi$ for this steer angle, including its sign.

   $\Delta\phi$ =

   (b) Compute the commanded phase of element 5 from $\phi_n = n\Delta\phi$, before any wrapping.

   ϕ_5 =

   (c) Give the value written to element 5 after wrapping into 0° to 360°.

   ϕ_5 =

   (d) Wrapping changed the number by two whole turns and did not change the pattern. Explain why in one or two sentences, and state what information the wrapped table no longer carries.

3. **The steered pattern.** The array is steered to $\theta_0 = 50^\circ$. Use $\psi = kd\,(\sin(\theta) - \sin(\theta_0))$ with $kd = 173.2^\circ$, and recall $\lambda/Nd = 29.1/112 = 0.260$.

   (a) Compute the array-factor argument ψ at the observation angle $\theta = 30^\circ$.

   ψ = [ ]

   (b) Locate the first null on the broadside side of the main lobe from $\sin(\theta) = \sin(\theta_0) - \lambda/Nd$.

   θ = [ ]

   (c) The matching null on the other side requires $\sin(\theta) = 0.766 + 0.260 = 1.026$. Explain what that result means physically and what the pattern looks like on that side of the beam.

   (d) Two commanded angles differ by 5° near broadside, and two others differ by 5° near 60°. Which pair requires the larger change in $\Delta\phi$, and what does that mean for an array searching a sector in equal phase steps?

4. **Beam broadening and scan loss.** Use $\theta_{\text{HP}} \approx 0.886\lambda/\left(Nd\cos\left(\theta_0\right)\right)$ and the broadside uniform-array directivity $D \approx 2Nd/\lambda$, continuing with $N = 8$, $d = 14$ mm, and $\lambda = 29.1$ mm.

   (a) Compute the broadside half-power beamwidth of the array.

   θ_{HP} = [ ]

   (b) Compute the half-power beamwidth when the beam is steered to $\theta_0 = 55^\circ$.

   θ_{HP} = [ ]

   (c) Compute the broadside directivity of the array in dBi. Then, using the fact that the effective aperture shrinks to its projection $Nd\cos\left(\theta_0\right)$ when the beam is steered, compute the expected gain reduction at $\theta_0 = 55^\circ$ and name the effect.

   $D\left(0\right)$ = [ ]

   ΔG = [ ]

   (d) A tracking requirement allows the beam to broaden to at most 20°. Find the largest usable scan angle, and state the trade-off this puts on the system design in one sentence.

   θ_0 = [ ]

5. **The inverse problem.** You read the following phase settings out of the beamformer. The array is the course array, $kd = 173.2^\circ$.

n	0	1	2	3	4	5	6	7
ϕ_n (deg)	0	260.7	161.4	62.1	322.8	223.5	124.2	24.9

(a) Difference neighboring elements, unwrap, and report the common step and the value of $\Delta\phi$.

$\Delta\phi =$ ____

(b) Recover the commanded steer angle.

$\theta_0 =$ ____

(c) Suppose the seven differences had come out as -99.3° for six of them and -84° for one. State two things that could produce that result, and what you would check first.

(d) The ADAR1000 sets phase on a 2.8125° grid. Give the phase element 1 takes on that grid and the resulting error.

$\phi_1 =$ ____

Documentation: _____